

MANAGEMENT OF CONSTIPATION

1. NORMAL STOOLS

PATIENT	DESCRIPTION	TYPICAL FREQUENCY
Breastfed	Seedy, loose stools	Once a week – 8x/day
Formula-fed	Pasty stools, firmer than breastmilk stools (Consistency no firmer than peanut butter)	Once every 4 days – 8x/day
> 1-year-old	Bristol Type 3 or 4	3 – 4x/week – 1x/day

2. BRISTOL STOOL CHART

TYPE	DESCRIPTION		
Type 1		Separate hard lumps, like “rabbit droppings”	Indicative of constipation
Type 2		Sausage-shaped but lumpy (T1 lumps stuck together)	
Type 3		Sausage-shaped with cracks on its surface	Normal, acceptable stool
Type 4		Sausage-shaped, smooth & soft	
Type 5		Soft blobs with clear-cut edges (passed easily)	Indicative of loose stools
Type 6		Fluffy pieces with ragged edges, a mushy stool	
Type 7		Watery, no solid pieces (Entirely liquid)	

3. DEFINITIONS

CLASSIFICATION	DEFINITION
Chronic Constipation	Constipation lasting > 8 weeks
Intractable Constipation	Constipation that does not respond to sustained optimal medical management

4. PRESENTATION

CLINICAL FEATURES	FINDINGS
Stool pattern	Less than 3 complete stools (Bristol Type 3 or 4) a week “Rabbit droppings” Bristol Type 1 stools Hard large stool (that may be difficult to flush)
Associated symptoms	Poor appetite that improves after passing large stools Colicky periumbilical or lower abdominal pain relieved after defecation, exacerbated by oral intake (2' gastro-colic reflex)
Symptoms with defecation	Distress / straining with stooling (with the exception for infants < 9 months – see “Infantile Dyschezia” below) Bleeding/anal pain associated with hard stools. Retentive posturing - Straight-legged, tiptoed, back-arching posture. - To avoid defecation, active contraction of pelvic floor muscles which fatigue → use gluteal muscles to squeeze the buttocks together
Fecal impaction (more common in age > 1-years-old)	Fecal mass felt in lower abdomen. Fecal loading in distal bowels seen on AXR. Spurious diarrhea (formless & very smelly; can be <u>thick & sticky</u> or <u>dry & flaky</u> in appearance) Fecal incontinence (staining underwear). Distended rectum → Reduced gastric emptying (neg feedback) → reduced appetite, nausea & vomiting. Symptoms reverse after passage of large stools.
Uncommon presentation	<u>Lower Urinary Tract Symptoms</u> - Lower abdominal pain while passing urine is mistaken for dysuria. - Distended rectum pushes on the bladder resulting in urinary frequency. - No fever, UC10/UFE negative.

4.1 INFANTILE DYSCHYZIA

- Definition:
 - Infants < 9 months of age
 - No red flag signs/symptoms
 - Straining & crying for ≥ 10 minutes before successful or unsuccessful passage of soft stools
- Developmental skill for coordinating voluntary/involuntary movements to expel stools not acquired.
- Treatment: reassurance & follow up. Glycerin suppositories are not advised as infants may be conditioned to wait for rectal stimulation to defecate.

5. PITFALLS IN HISTORY TAKING

- Have caregivers further elaborate what is meant by “normal” stools.
- Daily BO ≠ No constipation.
- Parent may not be the best historian – speak to the main caregiver or the patient directly

6. PATHOLOGICAL CAUSES OF CONSTIPATION

	CAUSE OF CONSTIPATION	RED FLAGS
Intestinal	Hirschsprung’s disease Ano-rectal malformations Neuronal intestinal dysplasia	Delayed passage of meconium > 48H Constipation starting < 1 month of life. Ribbon stools Abdominal distension + vomiting
Neurological	Spinal dysraphism Spinal cord trauma Neurofibromatosis	Absent cremasteric reflex, abnormal lower limb neurology Herald sign, Sacral dimple, Gluteal cleft deviation
Metabolic/ Endocrine	Hypothyroidism Hypercalcemia, Hypokalemia Vitamin D intoxication Diabetes Mellitus (late complication)	Signs & symptoms of hypothyroidism. Failure to thrive.
Drugs	Opioids Anticholinergic drugs (e.g., morphine, 1 st generation antihistamines, prochlorperazine, TCA)	
Others	Sexual abuse Anorexia nervosa Dietary protein allergy Cystic fibrosis	Bruising around anus. Failure to thrive.

7. INVESTIGATIONS

- There are no routine investigations for the diagnosis of functional constipation.
- Consider an XR abdomen to evaluate for fecal impaction:
 - An uncooperative patient with difficult abdominal examination.
 - No apparent clinical features of constipation on history & examination in a patient with abdominal pain and initial investigations return unremarkable.

8. PATHOPHYSIOLOGY OF FUNCTIONAL CONSTIPATION

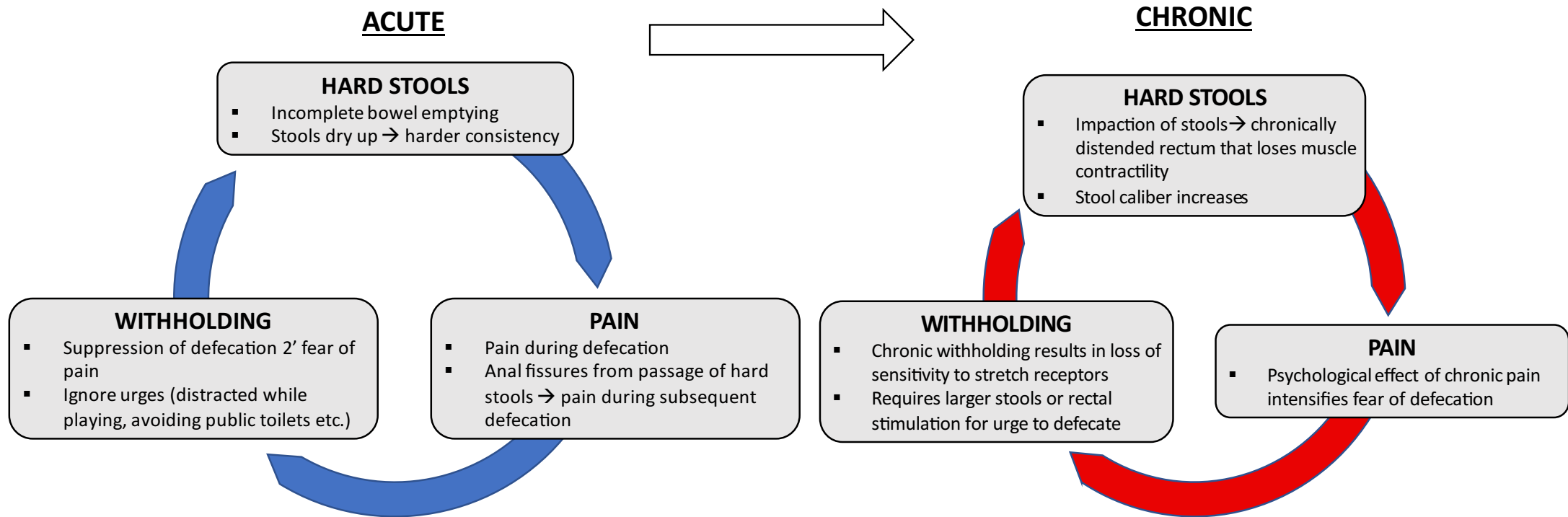


Figure 1: Identify patients caught in the vicious cycle of acute constipation & intervene to break the spiral towards chronic constipation.

9. MANAGEMENT

Do not use dietary interventions alone as first-line treatment for idiopathic constipation.

9.1 DISIMPACTION THERAPY

- Necessary before maintenance therapy for patients with impacted stools (e.g., history of spurious diarrhea, stool incontinence, fecal mass on palpation)
- Disimpaction is achieved when bowel output is mostly brown water with very few solid bits
- Oral method is preferred over rectal (for patients with chronic constipation)
- Adverse effects of rectal:
 - Intense abdominal cramps following rectal medication
 - Pain from anal fissures during insertion of fleet or passage of large, hard stools
 - Exacerbates child's overall negative psychological experience with defecation → withholding behavior
- Consider rectal if:
 - Patient's symptoms affect function
 - Presenting with vomiting (affects consumption of forlax)
- Adverse effects of oral:
 - Exacerbation of spurious diarrhea, stool incontinence; experience colic
 - Issues with compliance to medication in view of significant reconstituted volumes

9.1.1 DOSE

ORAL	AGE	NO. OF SACHETS PER DAY						
FORLAX® PEG – polyethylene glycol (Age > 2 years)		D1	D2	D3	D4	D5	D6	D7
	2 – 5 years	1	2	2	3	3	4	4
	6 – 11 years	2	3	4	5	6	6	6
	≥ 12 years	8	8	8	-	-	-	-
Instructions on prescription: - For disimpaction - Dissolve each sachet in at least 120ml of water - 8 sachets can be dissolved in 1L of water to be consumed within 12H (keep refrigerated)								
RECTAL*		AGE		DOSE (once, every 24H)				
SODIUM PHOSPHATE ENEMA (FLEET®)	Paediatric (66ml)	2 – 4 years		½ Paeds Fleet				
		5 – 11 years		1 Paeds Fleet				
	Adult (133ml)	≥ 12 years		1 bottle				
GLYCEROL SUPPOSITORY (2G)		1 – 11 months		1g (½ suppository)				
		1 – 2 years		2g				

**If macrogols are not available or patient is unable to consume macrogol/lactulose, consider rectal disimpaction followed by maintenance therapy.*

9.2 MAINTENANCE THERAPY

- Estimated duration of treatment = duration of untreated constipation
- Continued for at least 4 weeks after **regular bowel habits** have been established.
- Regular bowel habits:
 - Daily BO with complete bowel emptying (or at least 3 - 4 per week)
 - No symptoms of constipation (Bristol T3 or 4 stools, no straining, no pain, no features of impacted stools)
- Patients who are NOT toilet trained should remain on laxatives until toilet training is well established.
 - Voluntary bowel emptying is a necessary skill for regular bowel habits to be established
- Maintenance should be slowly weaned over a few months, titrated according to patient's symptoms and not abruptly discontinued
- Common pitfalls:
 - Premature discontinuation of maintenance therapy
 - Abrupt discontinuation of maintenance therapy
- Titration of laxatives:
 - Stimulant: Frequency of BO and/or urge experienced
 - Osmotic: Consistency of stools (Bristol Chart)

9.2.1 DOSE

DRUG		AGE	DOSE	ONSET
OSMOTIC LAXATIVES				
Lactulose		> 1 mo	1 – 3 ml/kg/day, in 2 divided doses	1 – 2 days
Forlax (Macrogol 4000) <i>(Each sachet dissolved in at least 100ml of water)</i>		2 – 5y	0.5 sachet daily	1 – 4 days
		6 – 11y	1 sachet daily	
		≥ 12y	1 – 3 sachets daily	
STIMULANT LAXATIVES				
Senna (Sennosides) 1 tab = 7.5 mg		2 – 3y	0.5 – 1 tablet OD	8 – 12H
		4 – 5y	0.5 – 2 tablets OD	
		6 – 17y	1 – 4 tablets OD	
Bisacodyl	Oral	4 – 17y	5 – 20 mg ON	10 – 12H
	Rectal	2 – 10y	5 – 10 mg OD	20 – 60 mins

9.3 BEHAVIOR MODIFICATION

9.3.1 LIFESTYLE CHANGES

- Ensure appropriate fluid intake

Category	Age	Water From Drinks Per Day*
Infant	7 months to 1 year	600ml
Child	1 to 2 years	900ml
Pre-school	3 to 6 years	1000 – 1200ml
Primary	7 to 12 years	(M) 1800ml; (F) 1600ml
Secondary / Junior college	13 to < 18 years	(M) 2600ml; (F) 1800ml

- Encouraging a balanced diet.
 - There is no evidence that increasing fiber-rich foods is effective in treating constipation.
 - Excessive fiber intake may worsen symptoms and increase soiling.
- Increasing physical activity to promote gut motility
- Providing foot support with a stool for young children whose feet cannot touch the floor

9.3.2 OVERCOMING WITHHOLDING BEHAVIOR

9.3.2.1 Common Triggers for Withholding Behavior

Age	Event	Description
≥ 6 months	Weaning	Introduction of formula milk from a completely breastfed child; weaning to solids
2 – 3 years	Toilet training	Some children develop fear of defecation during toilet training
4 – 7 years	Entering school	Change in routine; Fear of public toilets
Any age	Acute illness	10% drop in fluid intake can result in hard stools
	Travel	Change in routine, ignoring urges

9.3.2.2 Scheduled Toilet Sits

- Timed to occur after meals to take advantage of the gastro-colic reflex (or to occur during times patients typically experience fecal incontinence)
- Duration not more than 10 – 15 minutes to improve compliance.
- Calm, stress-free (no reprimanding child) and distraction-free (e.g., iPad, YouTube etc.)
- Complemented with positive reinforcement (e.g., star chart)

9.3.2.3 Topical Analgesia

- For patients with anal fissures
- Lignocaine 2% gel, apply thinly, TDS-PRN

10. FAQ (EDUCATION FOR CAREGIVERS)

- Parental misconceptions regarding oral medications
 - Daily use is not healthy for child – fear of “dependence” or “addiction”
 - Not as effective as rectal medications – to correct previous wrong practices and expectations regarding oral medications
 - Frequent use of rectal suppositories → patient relies on medication for urge to BO
- Managing expectations
 - Duration of therapy is proportionate to the chronicity of constipation (if duration of constipation = 3 months before seeking help, estimated duration of therapy to achieve regular stools = 3 months at least)
 - Medications can result in abdominal cramps, bloating and/or worsen fecal incontinence especially when initiating therapy
 - Having 1 day of normal stool is not indication to stop maintenance therapy – symptoms of constipation should have resolved for 1 month before considering discontinuation
 - Medication should not be abruptly discontinued – but slowly tapered

11. FOLLOW UP

- All patients receiving disimpaction should be reviewed in 1 – 2 weeks so that medications can be titrated if the current regime does not work.

12. REFERENCES

1. **NICE Guidelines.** Constipation in children and young people: diagnosis and management. National Institute for Health and Clinical Excellence (NICE), 2010. Available at: www.nice.org.uk/guidance/cg99.
2. **NASPGHAN and ESPGHAN.** Evaluation and Treatment of Functional Constipation in Infants and Children: Evidence-Based Recommendations. Journal of Pediatric Gastroenterology and Nutrition, 2014; 58(2): 258–274. DOI: 10.1097/MPG.0000000000000266.
3. **Singapore Medical Journal (SMJ).** Ho J.M.D., How C.H. Chronic constipation in infants and children. Singapore Med J 2020; 61(2): 63–68. DOI: 10.11622/smedj.2020014.